CMSC216: Bonus Review 2A

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Bonus Review Rules

- 2 Questions will be shown with about 5min per question, 10min total, time limit enforced on Gradescope Quiz
- Each correct answer worth 1 Engagement Point
- Cooperation is allowed and encouraged within your discussion section - *talk to your neighbors, form a class consensus!*
- Questions and Scores posted after all sections have completed bonus review, likely by the following day
- Review is Open Resource just like the exam: https://www.cs.umd.edu/~profk/216/exam-rules.pdf

Staging

- Open up the Gradescope Review Quiz for the day
- Once started, the quiz closes after 10min

Get everybody and their stuff together

Okay...



Question 1

```c
int quotient_greater(int numer,
                     int denom,
                     int thresh)
{
  int quot = numer / denom;
  if(quot > thresh){
    return 1;
  }
  else{
    return 0;
  }
}
```

```
.global quotient_greater
quotient_greater:
        movl    %edi, %eax
        cltd
        idivl   %esi
        cmpl    %edx, %eax
        jle     .BELOWTHRESH
        movl    $1, %eax
        ret
.BELOWTHRESH:
        movl    $0, %eax
        ret

```

Above is a correct C function and a BUGGY assembly implementation. Which of the below best describes the problem in assembly and its fix?

- (A) Sign extension in preparation for the division is not correct; use the `cwtd` instruction
- (B) The argument registers are used incorrectly; change the first line to `movl (%edi),%eax`
- (C) The `%esi` register cannot be used in division; copy `%esi` to a different register and use that one in division.
- (D) The argument in `%edx` is changed during division; copy `%edx` to a different register early then compare against that one later.

Question 2

Consider this struct and function prototype.

```
// packed_struct_main.c
typedef struct {
  short first;
  short second;
} twoshort_t;

short sub_struct(twoshort_t ti);
```

Which of the nearby instruction sequences will set the DX-family register to be the value of `ti.second` if placed at beginning `sub_struct()`?

```
sub_struct:
  # (A) edx = ti.second;
  movq %rdi, %rdx
  sarl $8, %edx
  andl $0xFF, %edx

  # (B) edx = ti.second;
  movl %edi, %edx
  sall $16, %edx
  andl $0xFFFF, %edx

  # (C) edx = ti.second;
  movl %edi, %edx
  sarl $16, %edx
  andl $0xFFFF, %edx

  # (D) edx = ti.second;
  movl $0, %edx
  movw 2(%rdi), %dx

  # (E) edx = ti.second;
  movzwq (%rdi), %dx

  # (F) edx = ti.second;
  xorq %rdx,%rdx
  movswq 2(%rdi), %dx
```